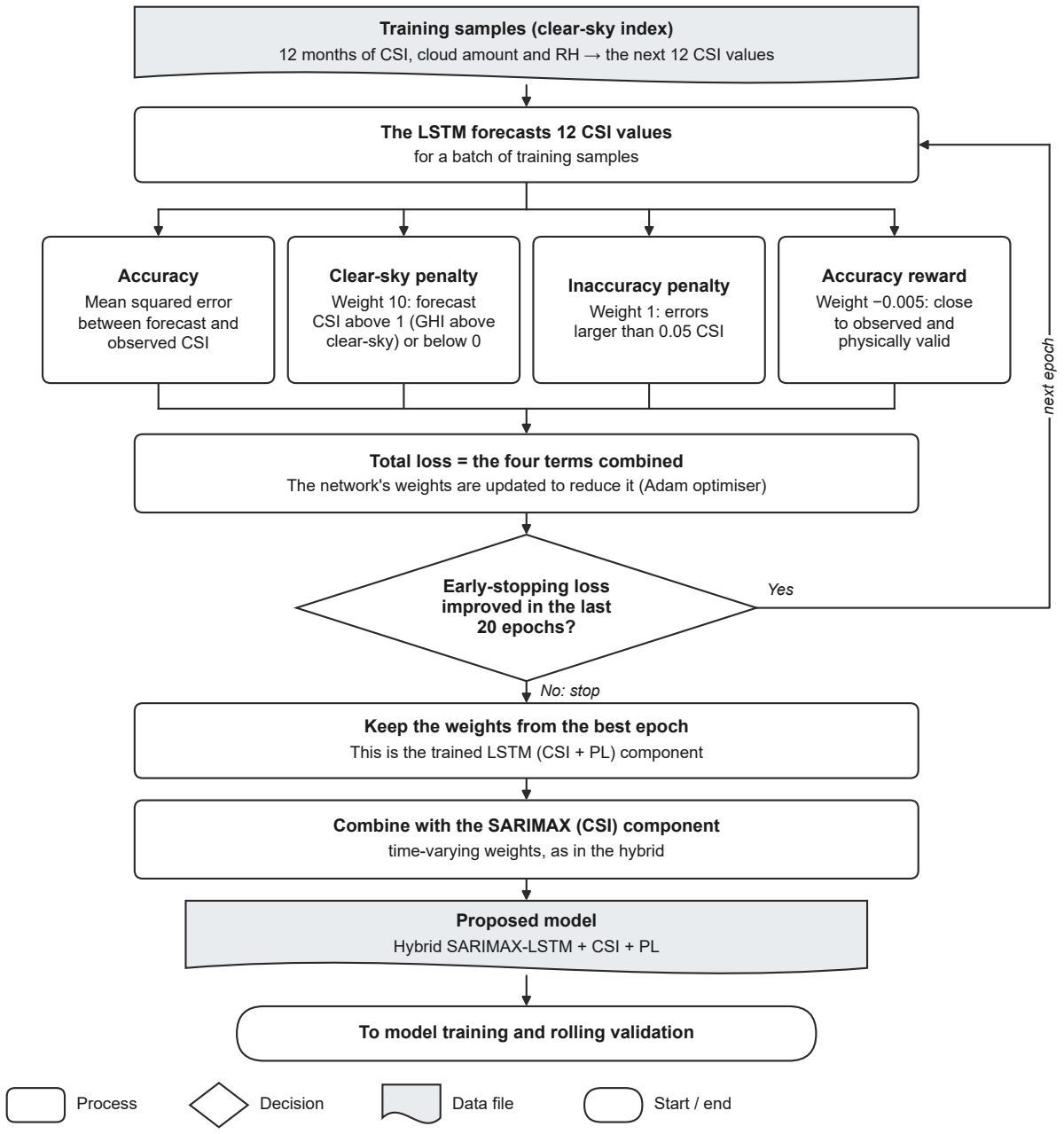

Training samples (clear-sky index)
12 months of CSI, cloud amount and RH → the next 12 CSI values
The LSTM forecasts 12 CSI values
for a batch of training samples
Accuracy
Mean squared error between forecast and observed CSI
Clear-sky penalty
Weight 10: forecast CSI above 1 (GHI above clear-sky) or below 0
Inaccuracy penalty
Weight 1: errors larger than 0.05 CSI
Accuracy reward
Weight −0.005: close to observed and physically valid
next epoch
Total loss = the four terms combined
The network's weights are updated to reduce it (Adam optimiser)
Early-stopping loss improved in the last 20 epochs?
Yes
No: stop
Keep the weights from the best epoch
This is the trained LSTM (CSI + PL) component
Combine with the SARIMAX (CSI) component
time-varying weights, as in the hybrid
Proposed model
Hybrid SARIMAX-LSTM + CSI + PL
To model training and rolling validation
Process
Decision
Data file
Start / end